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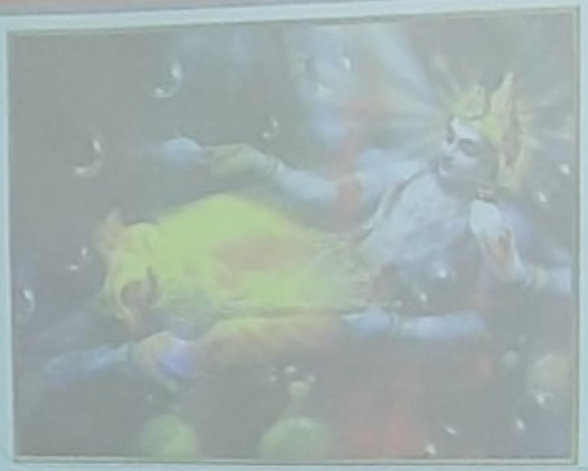
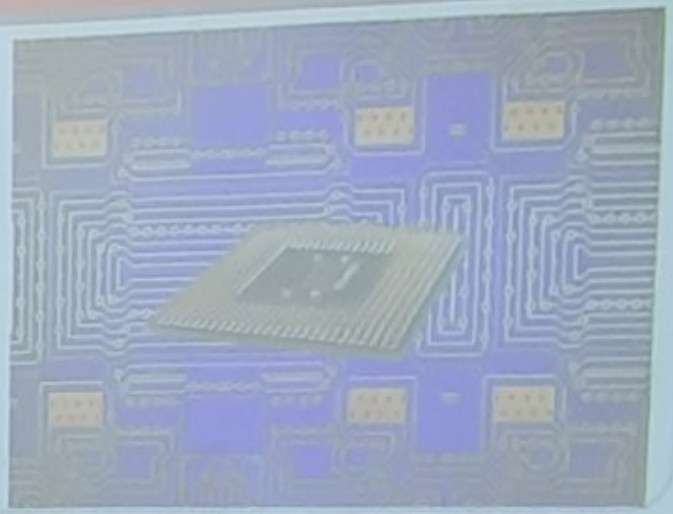
CPU Control Unit





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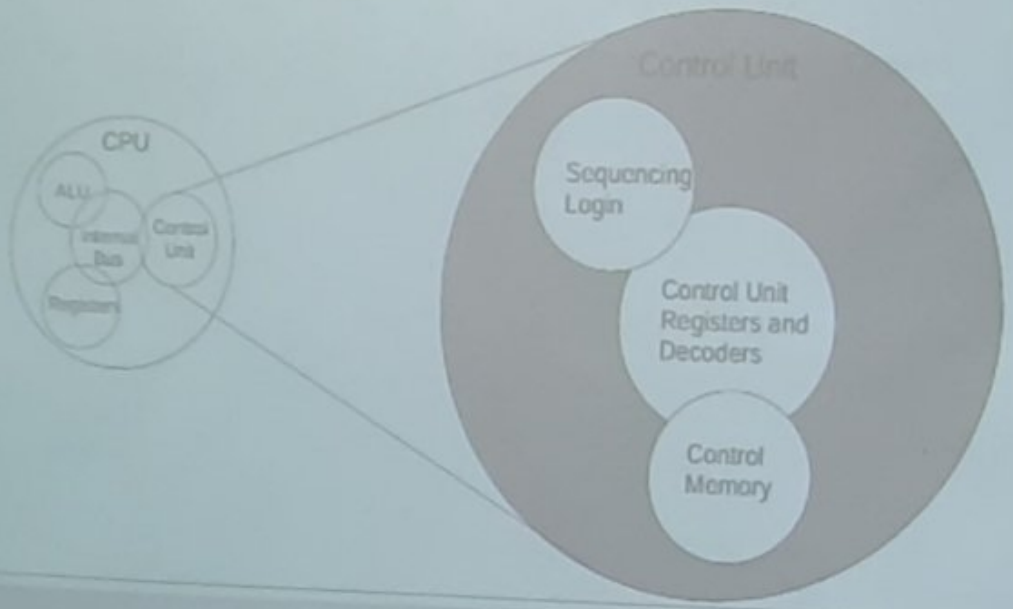
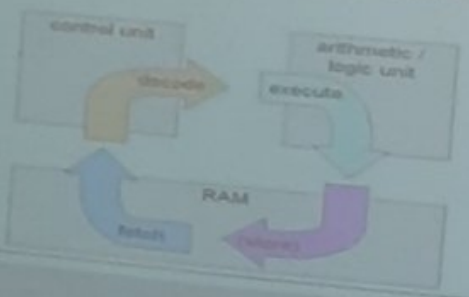
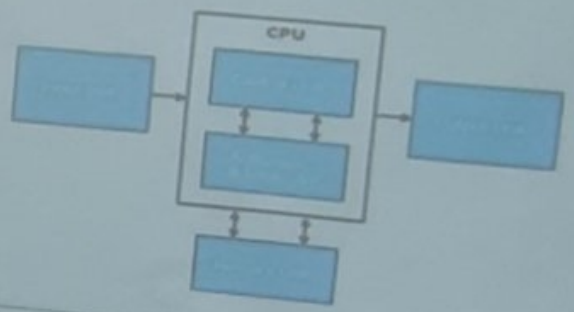
The Control Unit Structure



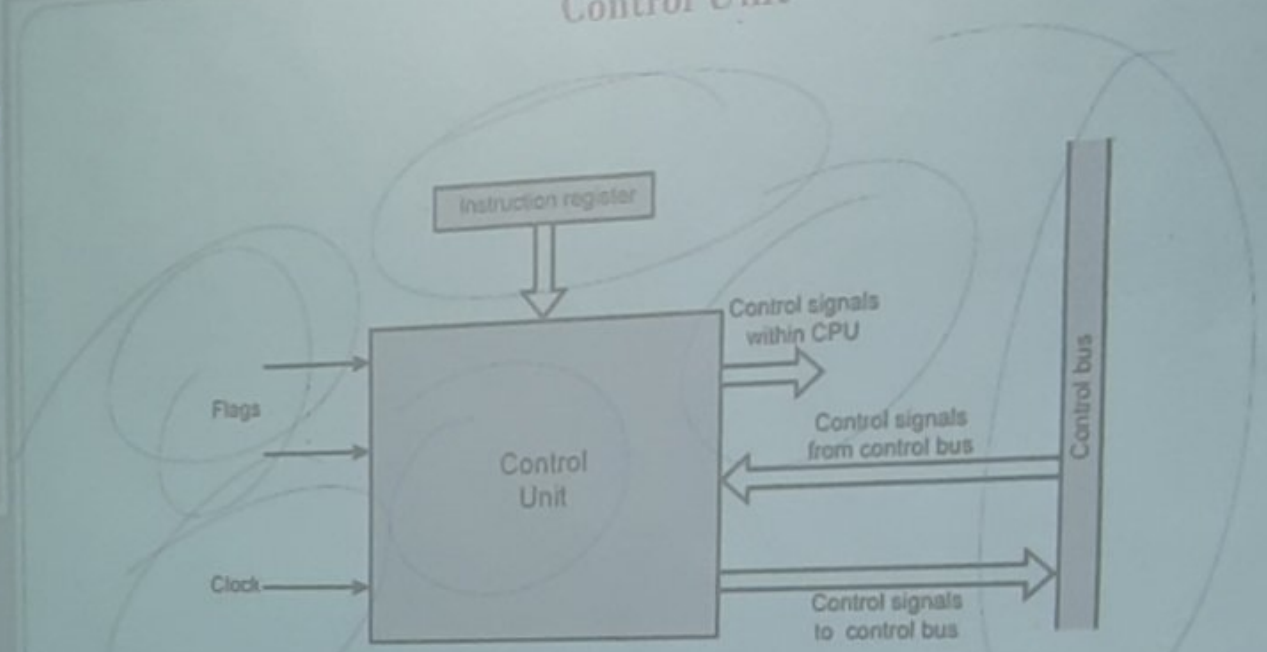
Control Unit

- *A control unit, or CU, is circuitry within a computer's processor that directs operations.*
- *The CPU control unit is a component of the computer's CPU (central processing unit) that directs the processor's operation.*
- It instructs the memory, logic unit, and both output and input devices of the computer on how to respond to the program's instructions.
- The control unit fetches internal program instructions from the main memory to the processor instruction register, and it generates a control signal based on the contents of this register to supervise the execution of these instructions.
- The Control Unit can be thought of as the brain of the CPU itself.
- It controls based on the instructions it decodes, how other parts of the CPU and in turn, rest of the computer systems should work in order that the instruction gets executed in a correct manner.
- CPUs and GPUs are examples of devices that use control units





Control Unit



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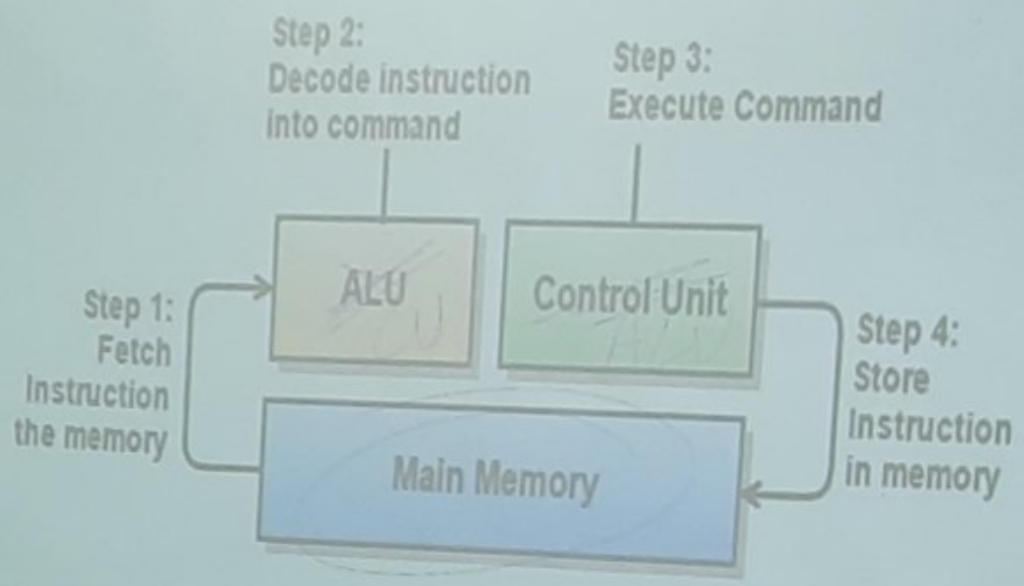
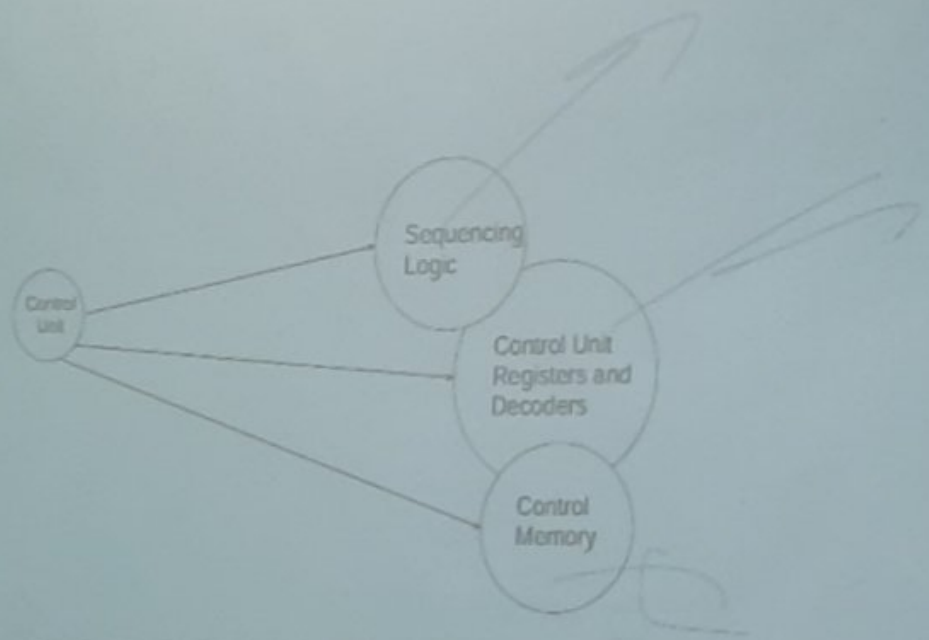
Control Unit Workings



Functions of Control Unit

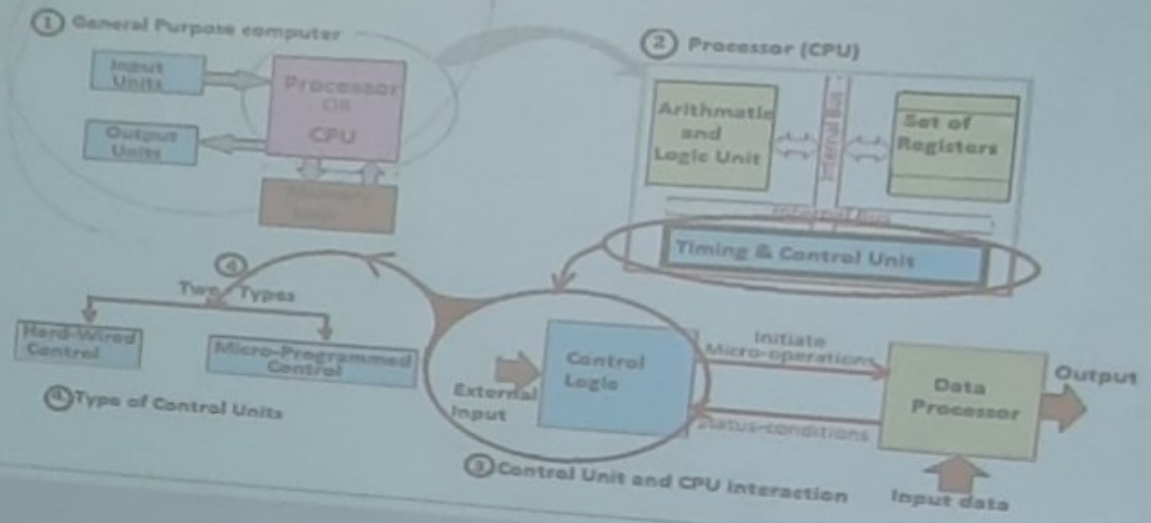
- It *coordinates* the sequence of data movements into, out of, and between a processor's many sub-units.
- It *interprets* instructions.
- It *receives* external instructions or commands to which it converts to sequence of control signals.
- It *controls* data flow inside the processor.
- It *controls* many execution units (i.e., ALU, data buffers and registers) contained within a CPU.
- It also handles multiple tasks, such as fetching, decoding, execution handling and storing results.

Control Unit Components



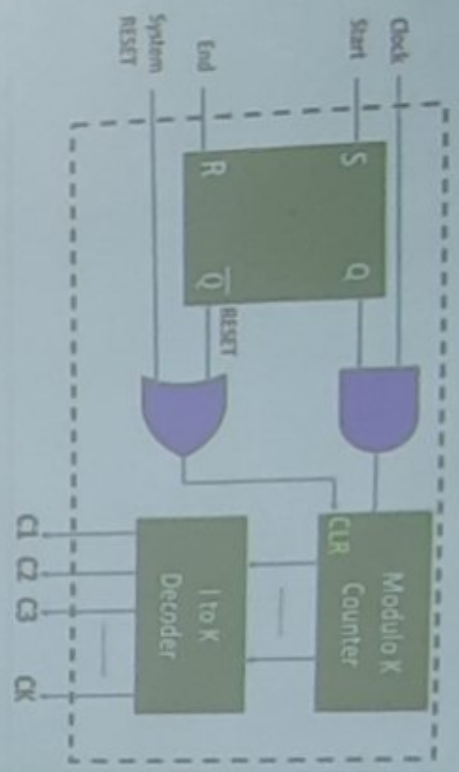
Classification of Control Unit

- There are two types of control units.
  A. *Hardwired* control unit
  B. *Microprogrammed* control unit.



Hardwired Control Unit

- *The control signals that are important for instruction execution control are generated by specially designed hardware logical circuits.*
- *It is known as a hardwired controller.*
- One can not modify the signal generation method without physical change (flip-flops, gates, etc.) of the circuit structure.
- *A hardwired control is a method of generating control signals with the help of Finite State Machines (FSM).*
- The control hardware can be viewed as a state machine .
- The outputs of the state machine are the control signals.
- The sequence of the operation carried out by this machine is determined by the wiring of the logic elements and hence named "hardwired".
- ***RISC** architecture is based on the hardwired control unit.*



Hardwired Control Unit

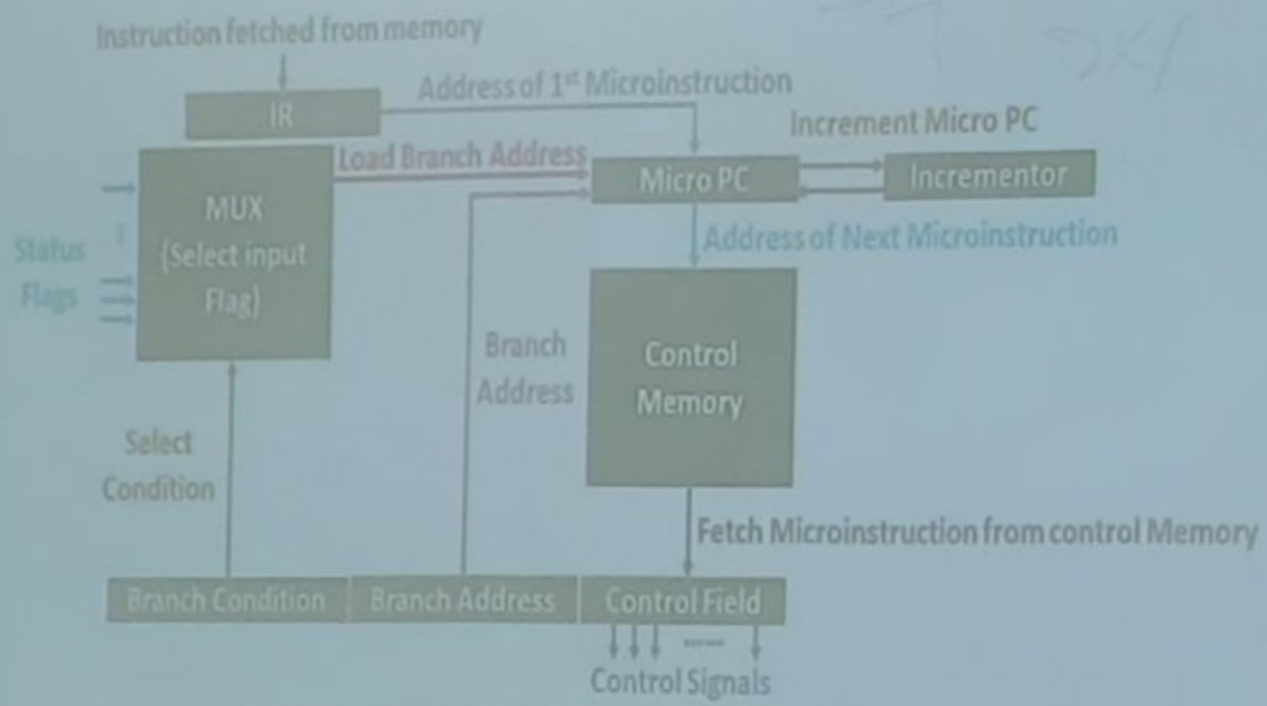
- Advantages of Hardwired Control Unit:
  1. Fast because control signals are generated by combinational circuits.
  2. The delay in generation of control signals depends upon the number of gates.

- Disadvantages of Hardwired Control Unit:
  1. More the control signals required by CPU, more complex will be the design of control unit.
  2. No Flexibility. Modification in control signal is very difficult i.e. it requires rearranging of wires in the hardware circuit.
  3. Difficult to add new feature in existing design of control unit.

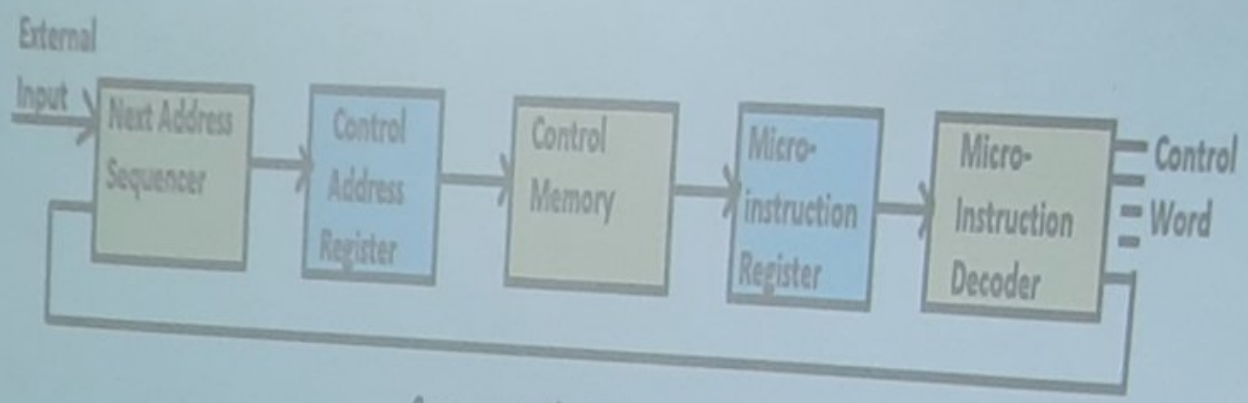
Microprogrammed Control Unit

- *A microprogrammed control unit is a control unit that saves binary control values as words in memory.*
- A controller results in the instructions to be implemented by constructing a definite collection of signals at each system clock beat.
- Each of these output signals generates one micro-operation including register transfer.
- Thus, the sets of control signals are generated definite micro-operations that can be saved in the memory.
- By creating a certain collection of signals at every system clock beat, a controller generates the instructions to be executed.
- Each bit in the microinstruction is connected to a single control signal.
- The control signal is active when the bit is set., and it becomes inactive when it is cleared.
- A sequence of these microinstructions can be kept in the internal 'control' memory.
- ***A microprogram-controlled computer's control unit is a computer within a computer.***

Microprogrammed Control Unit



Organization of Microprogrammed Control Unit



Components of a MPU

Microprogrammed Control Unit

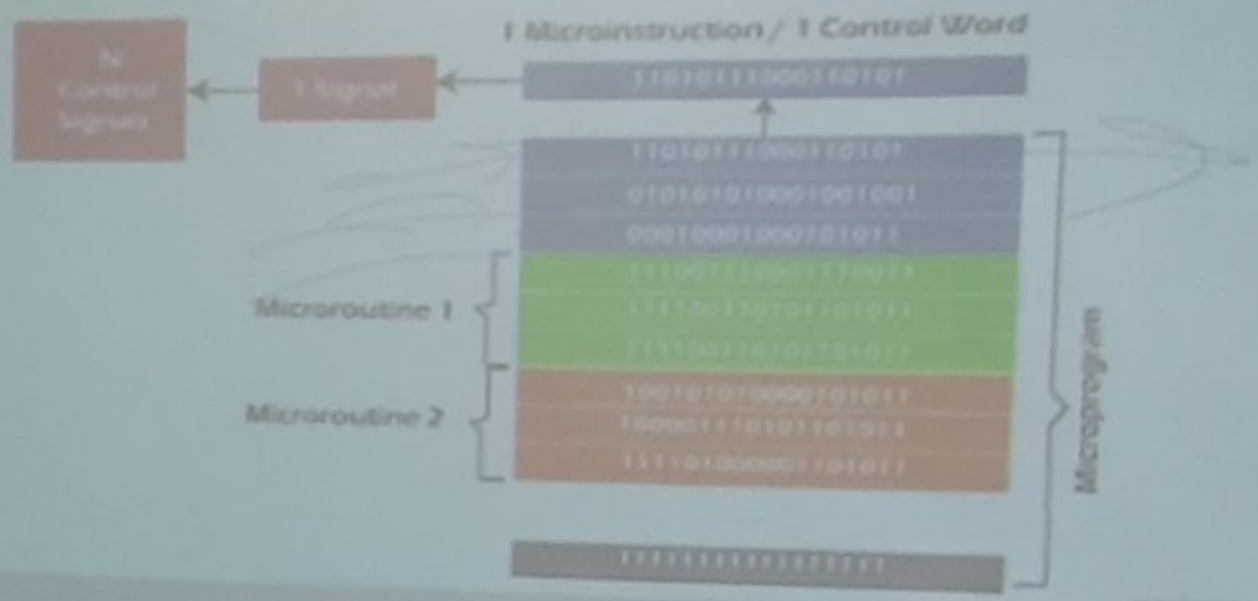
- **Advantages**
  - It is a more systematic design of the control unit.
  - It is simpler to debug and change.
  - It can retain the underlying structure of the control function.
  - It can make the design of the control unit much simpler. Hence, it is inexpensive and less error-prone.
  - It is used to control functions implemented in software and not hardware.
  - It is more flexible.
  - It is used to complex function is carried out easily.
- **Disadvantages**
  - Adaptability is obtained at more cost.
  - It is slower than a hardwired control unit.

Related Terminologies

- *Control Word:* A control word is a word whose individual bits represent various control signals.
- *Micro-routine:* A sequence of control words corresponding to the control sequence of a machine instruction constitutes the micro-routine for that instruction.
- *Micro-instruction: Individual control words in this micro-routine are referred to as microinstructions.*
- *Micro-program:* A sequence of micro-instructions is called a micro-program, which is stored in a ROM or RAM called a Control Memory (CM).
- *Control Store:* the micro-routines for all instructions in the instruction set of a computer are stored in a special memory called the Control Store.

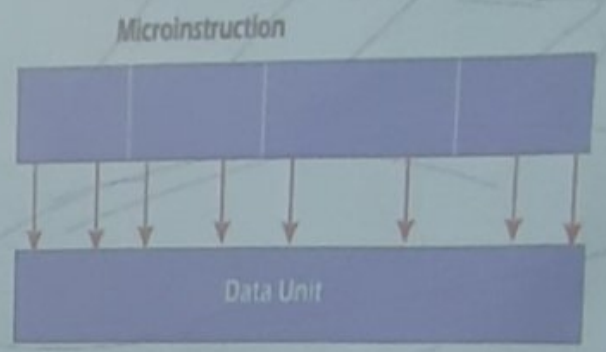
Microprogrammed Control Unit: Classification

- *Horizontal Micro-programmed Control Unit*
- *Vertical Micro-programmed Control Unit*



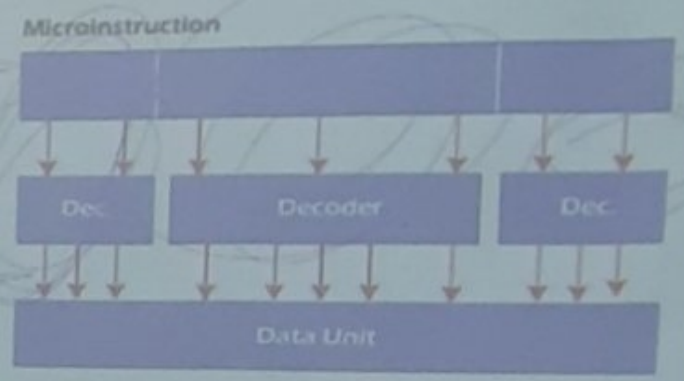
Horizontal Microprogrammed Control Unit

- *The control signals are represented in the decoded binary format that is 1 bit/CS.*
- Example: If 53 Control signals are present in the processor then 53 bits are required.
- More than 1 control signal can be enabled at a time.
- It supports longer control words.
- It is used in parallel processing applications.
- It allows a higher degree of parallelism.
  - *If degree is n, n CS is enabled at a time.*
- It requires no additional hardware(decoders).
- It means it is faster than Vertical Microprogrammed.
- It is more flexible than vertical microprogrammed



Vertical Microprogrammed Control Unit

- *The control signals are represented in the encoded binary format.*
- For N control signals- Log2(N) bits are required.
- It supports shorter control words.
- It supports easy implementation of new control signals therefore it is more flexible.
- It allows a low degree of parallelism i.e., the degree of parallelism is either 0 or 1.
- Requires additional hardware (decoders) to generate control signals, it implies it is slower than horizontal



	Hardwired Control Unit	Micro-programmed Control Unit
Implementation	Fixed set of logic gates and circuits	Microcode stored in memory
Flexibility	Less flexible, difficult to modify	More flexible, easier to modify
Instruction Set	Supports limited instruction sets	Supports complex instruction sets
Complexity of Design	Simple design, easy to implement	Complex design, more difficult to implement
Speed	Fast operation	Slower operation due to microcode decoding
Debugging and Testing	Difficult to debug and test	Easier to debug and test
Size and Cost	Smaller size, lower cost	Larger size, higher cost
Maintenance and Upgradability	Difficult to upgrade and maintain	Easier to upgrade and maintain

Exercise

- Assume 4k words, which are supported with the help of hypothetical control unit. There are total 16 flags and 64 control units in the hardware. Now find the size of control memory in bytes and the size of control word in bits for two types of programming, i.e., Horizontal and Vertical programming.

Branch Condition	Flag	Control Field	Control Memory address